A blind steganalysis-based predictive analytics of numeric image descriptors for digital forensics with Random Forest & SqueezeNet

Wasiu Akanji
Department of Computer Science, Lagos State University of Science & Technology
Lagos, Nigeria
0000-0001-8093-1822

Ogobuchi Okey
Department of Systems Engineering & Automation
Universidade Federal de Lavras
MG, Brazil
0000-0003-0686-2763

Saheed Adelanwa
Department of Computer Science, Lagos State University of Science & Technology
Lagos, Nigeria
0000-0002-2066-5347

Oluwafunsho Odesanya
Department of Computer Science, Lagos State University of Science & Technology
Lagos, Nigeria
0000-0002-0383-2035

Taiwo Olaleye
Department of Computer Science, Federal University of Agriculture
Abeokuta, Nigeria
0000-0001-6222-7575

Mary Amusu
Department of Computer Science, Lagos State University of Science & Technology
Lagos, Nigeria
0000-0003-2793-7584

Akinfolarin Akinrinlola
Department of Computer Science, Lagos State University of Science & Technology
Lagos, Nigeria
0000-0002-8046-554x

Abiodun Oladejo
Deaprtment of Electrical/Electronics Engineering,
Ladoke Akintola University of Technology
Ogbomoso, Nigeria
0000-0002-2060-0957

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***Abstract*— Image steganalysis have been a prominent study in digital forensics and the data science use case of artificial intelligence has been widely adopted in conceptual frameworks. In existing studies, deep learners gain prominence for intrusion detection systems while other dissimilar modules are used for feature extraction. Hence, this study rather employs deep learners as image embedding networks aimed at feature extraction for a predictive analytics of image steganalysis. The extracted numeric image descriptors trains three learner algorithms for pattern recognition using a 10 fold cross-validation system. Experimental result indicates the ensemble of Random forest algorithm and SqueezeNet image embedder as the best for steganalysis in digital forensics while the size of the training set turns out to be insignificant for the supervised machine learning study.**

Keywords*— *Image steganalysis, Machine learning, Digital forensics, image embedding

I. Introduction

The branch of forensic science which focuses on identification, acquisition, processing and analysis of electronically-stored data continues to grow in lips and bounds owing to its crucial role in law enforcement initiatives. This is further buoyed by the fact that data is now generated in an unprecedented rate [1] and utilized by individuals for both legitimate and sinister intents in a data centered-economy that we are in. In law enforcement, electronic evidence is undoubtedly an acceptable component of criminal investigation [2] hence the importance of digital forensics especially for security purposes. For cyber security initiatives, digital forensics collects and analyzes computer-related proof in aiding network vulnerability mitigation and as well as counterintelligence investigations. Whereas data security is central to cyber security, discovery of hidden secretes in data through data mining is likewise germane for cyber security intentions [3]. This has become imperative due to the fact that digital revolution has triggered increase in data collection and transmission which is prone to manipulations. Data can easily conceal implicit information which conventional inspection cannot easily detect in what is referred to as data hiding. Data hiding indeed predates contemporary history for concealing messages [3]. The Greeks in antique time conceal messages by inscription of texts on wood surface covered with wax to prevent full disclosure. The intended recipient scrapes off the wax to obtain the information that is communicated. A native approach entails shaving of slave's head to engrave messages for transmission. After a while, grown hair conceals the message from unintended audience [2]. Similar approach is imbibed in informatics to hide confidential data through various encryption schemes which precludes intruders from full disclosure. Especially, steganography techniques provides an impeccable tool for data concealment, to facilitate hidden communication among undisclosed parties [4]. Majority of contemporary steganography systems employ multimedia objects including video, audio, image etc. as a cover to conceal secret information owing to the popularity of digital [5]. Image steganography is however common on digital platforms which reliably conceals information for malicious or legitimate intents. For malicious purposes, messages are concealed in digital images and transmitted between nodes on the internet which renders systems and networks vulnerable to attacks undetected. This has aided several cyber security breaches hence the need for continued improvements of digital forensic disruptive initiatives. However, such malicious concealment could be detected through data mining approaches in order to identify

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the incidence of shrouded information. Consequent upon the foregoing, the employment of deep learning for detecting steganography is popular in literature with state-of-the-arts but the image embedding approach is rare in steganalysis studies. Image embedding is a deep learning technique for feature extraction which could be served into a machine learning model for predictive analytics. Image embedders accepts image signal inputs and uses deep learning models to compute feature vectors for each input. A table of numeric image descriptors is subsequently outputted for further analysis. In other studies, the deep learner embedders are rather used as learning models unlike the vector-extraction approach of this study. The embedders include Inception v3 [6], VGG-16 [7], SqueezeNet [8], VGG-19 [9], etc., which are either trained on Kaggle's Painter by Number, Authors, ImageNet, and other prominent training sets. The aforementioned gap in literature thus motivates this study. Therefore, the objective of the study is to employ the embedders as feature extractors. Hence, rather than employing the deep embedders as learning models as is the practice in literature, this study deploys five of them for feature extraction purposes, which is a novel approach in predictive steganalysis. The extraction of numeric feature vectors is achieved through the transfer learning methodology [10] where the pre-trained five deep learners are used to extract the numeric vectors from both pure and stego images. The extracted vectors will be employed to train Support Vector Machine (SVM), Random Forest (RF), and Multilayer Perceptron (MLP) machine learning algorithms to detect the possibility of steganography in cover images (stego). Evaluation of the embedders is implemented through the instrumentality of 10-fold cross validation. The study therefore contributes to knowledge by pioneering the employment of image embedders as feature extractors for steganalysis, and the subsequent performance evaluation of machine learning algorithms on each of the extracted feature vectors through the interpretation of the performance metrics. The rest of the study is structured in the following format: related literature is reviewed in section II, while the conceptual framework of the study is presented in III. The result is discussed in IV, and the conclusion and recommendation is presented in section V.

II. RELATED STUDIES

First, This section discusses existing studies on steganalysis on one hand, especially with predictive analytics approach, and studies that employs predictive analytics on image descriptors extracted through deep embedding, on the other hand. The two research directions are narrowed to predictive modelling approaches in order to uniquely identify gaps in literature and subsequently consolidate on existing studies which have employed diverse conceptual frameworks in their methodologies. The investigation of deep learning network with respect to their scalability for steganography is the thrust of the study of [11]. The study employed Convolutional Neural Network (CNN) to investigate research questions including determining if the ranking of CNN variants would change in the area of accuracy and as well as if CNN will collapse when the database is larger than what is common in state-of-the-arts. The analysis of deep learning models for spatial image staganalysis achieved by Ye-Net, Xu-Net, SR-Net, Yedroudj-Net, GBRAS-Net, and Zhu-Net is conducted by [12]. Different image combinations with filter normalization ranges were likewise evaluated in the study with the accuracy performance metric evaluation tool. The study of [13] implements a SPAR-RL cost learning context with reinforcement learning which consolidates on existing studies. The methodology decomposes the cover image into pixel-wise actions towards maximizing rewards from a simulated steganalytic setting. The work of Saad, Mohamed, & Hafez [14] proposes a coverless concept to detect the presence of a secret message in a cover image using an ensemble of optical mark recognition with rule-based machine learning, while [15] employed deep learning based edge detection system for the proposed steganalysis. The performance evaluation of stego-image security in predictive analytics is the main thrust of the study conducted by [16] which seeks to intentionally misuse spatial data towards examining the dynamics of steganalysis. A multi-class blind steganalysis is conducted in the work of [17] using deep residual networks. The study categorized input data features into a binary class of cover or stego image instance in an attempt to resolve the problem of vanishing descent that impedes accuracy of machine learning models for detecting steganalysis. Similarly, a Secured Vector Machine is deployed for supervised machine learning for image security classification [3] on heart, satellite, shuttle, and diabetes data. A multi-filter and Squeeze-Convolutional Nueral Network for image steganalysis is studied in the work of [18] through the construction of a network with diverse filter modules with the squeeze-and-excitation modules which efficiently captures the intended embedding artifact. For other studies aside steganalysis that employed deep embedders for image feature extraction, diverse experimental results are returned. In [19], a dual phase feature extraction approach is implemented to identify most significant feature attributes from scanned image signals, comprising of positive and negative instances. The study trained Support Vector Machine classifier on two sets returning a 98.39% and 90.0% accuracy rates from the image embedding framework. In [20], the ResNet image embedding model is deployed to diagnose medical cases on chest X-ray set of 1078 images yielding 96% accuracy metric. Wang L. et al. deployed CNN for detection system with a 92.6% precision [21]. Their training set include of image signals and in a similar study to the approach of this study, Narin et al. employs deep learning embedding networks including ResNet50, Inception-ResNetV2 and ResNet50 on image inputs [22]. Whereas, a class decay technique is used in the work of [23] to identify suspected anomalies in image signals with feature extraction by image embedder. The conceptual framework can diagnose Covid-19 cases with 95.12% accuracy. Feature vectors from CXR images were mined via deep learning by [24] while a Support Vector Machine is trained on the input attributes for consequent Covid-19 analysis. Similarly, the work of [25] presents a distant domain transfer learning methodology on medical imaging. Unlabeled sets of images is used for a categorization precision of 96%, which is 13% better than the non-transfer embedding approaches. Image input set is used in the work of [26] which uses a deep learning model on COVNet training set, to classify similar medical conditions. The Operating Characteristic Curve, Sensitivity, and the Area Under the Curve (AUC) performance metrics evaluators shows the robustness of the conceptual framework. The transfer learning concept was the main thrust

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in [27] through deep embedders including VGG, AlexNet, GoogleNet, Xception, and ResNet. While the work of [28] entails eight-number models for the diagnosis of the infection through X-ray image signals using hyper-parameter tuning. An accuracy of 98.33% is returned by ResNet-34. As evident from the foregoing, studies have not evaluated the performance of deep image embedders as feature extractors for steganalysis which is germane towards consolidating on existing studies and as well as charting feature direction. Similarly, the work of [25] presents a distant domain transfer learning methodology on medical imaging. Unlabeled sets of images is used for a categorization precision of 96%, which is 13% better than the non-transfer embedding approaches. Image input set is used in the work of [26] which uses a deep learning model on COVNet training set, to classify similar medical conditions. The Operating Characteristic Curve, Sensitivity, and the Area Under the Curve (AUC) performance metrics evaluators shows the robustness of the conceptual framework. The transfer learning concept was the main thrust in [27] through deep embedders including VGG, AlexNet, GoogleNet, Xception, and ResNet. While the work of [28] entails eight-number models for the diagnosis of the infection through X-ray image signals using hyper-parameter tuning. An accuracy of 98.33% is returned by ResNet-34. As evident from the foregoing, studies have not evaluated the performance of deep image embedders as feature extractors for steganalysis which is germane towards consolidating on existing studies and as well as charting feature direction, which is the aim of this study.

III. RESEARCH METHODOLOGY

This study examines the expertise of image embedding networks on accurate detection of steganography in digital images. It seeks to establish the feature extraction suitability of the embedder networks for image steganalysis which is a complete departure from trend in literature. The study will further identify the best machine learning model fit for steganography predictive analytics. As could be observed from the conceptual framework in Figure 1. A three-phase conceptual framework is employed for the study which includes digital image acquisition; extraction of numeric image descriptors through the instrumentality of five image embedders (Squeezenet, InceptionV3, VGG16, VGG19, and Painters): which is the main thrust of this study; and steganalysis through machine learning. The public image signals (pure images and images with hidden messages-stego) acquired for digital forensics is fed into the image embedders simultaneously to extract the numeric image descriptors of unique sizes (relative to each embedder network). The machine learning models are fitted on the extracted numeric vectors for training and testing consecutively. The Python-based Orange data mining toolkit is used for the study simulation with 10 fold cross-validation methodology.

A. Phase I: Digital Image Acquisition

The image data used in this study are public signals encapsulated in two categories of cover image (original images without steganography) and embedded image (stego images with hidden secrets) which are captured for forensics [29]. The activity overview of the public set justifies its choice for this study. It attracts a 0.08 download per view ratio with 738 views and 56 total downloads so for steganalysis. A total of 4319 files are cover images (in JPG) while 4013 instances are embedded images (in JPG-embedded extension) making a total of 8332 image input signals for the study. Each of the 8332 files (pure and embedded) are fed into the five image embedding networks for extraction of numeric image descriptors, a resulting concatenation that makes up the labelled training set for machine learning.

B. Phase II: Feature Extraction

The extraction of numeric image descriptors takes place in this phase of the conceptual framework through image embedding networks. Image embedding for feature extraction entails submitting image signals into pre-trained deep learning models to get the vector representations of the inputs, which serves as the numeric description of the input image. The extraction is implemented through the instrumentality of transfer learning on dedicated digital server or on a local server. Upon successful extraction of the numeric descriptors from both clean and altered digital images, numeric vectors of the two versions are encapsulated and labelled according as *positive* (image with secret message) and *negative* (image signal without secret message). A description of each deep embedder is presented in Table 1 and described below:

- The deep neural network *Squeezenet* in this instance, returns 1000-no numeric vector attributes on each image instance which represents negative and positive pneumonia cases. The SqueezeNet is reputable to be a fast and small embedder for image recognition with far less computational overheads, compared to other deep emebdders [8]. The model is pretrained on the popular imageNet and attains AlexNet-level accuracy with 50x fewer parameters. It's fire module is the foundation of three stages of smaller network, which is achieved by swapping 3 × 3 filters with 1 × 1 filters. An input reduction is achieved on the remaining 3 × 3 filters and lastly the late down-sampling in the network

- InceptionV3 architecture is an improvement on V1, and V2. It incorporates batch normalization layers to streamline training, including the addition of factoring convolutions with larger spatial filters which greatly improves computational efficiency [30]. The main building block is the inception module with different dimensions of the kernel, 1x1 and 3x3. The network extracts 2048 feature vectors from each input image.

- VGG (16 and 19): The embedder family uses convolutional layers and the activation function rectified

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linear unit (ReLU). The VGG-16 and VGG-19 embedders has 16 and 19 layers respectively with a filter dimension 3x3 [31]. Both embedders extracts 4096 feature vectors from each signal input.

- Painters is a deep convolutional network embedder trained to predict painters from arkwork images. Its inclusion is premised on its ability for pattern recognition on artistic image signals [32] which could describe stego images in this context. Activations of the penultimate layer of the network is used for the embedding purpose. A total of 2048 feature attributes are extracted by the deep embedder from each input.

C. Phase III: Steganalysis through machine learning

Evaluation of the extracted feature vectors is done through predictive analytics by Support Vector Machine (SVM), Random Forest (RF) and Multilayer Perceptron (MLP) algorithms. The supervised machine learning approach is employed for the simulation where the three learner algorithms are initially trained with a portion of the training dataset (extracted numeric image descriptors) for pattern recognition and subsequently, the other portion is used for predicting the presence or otherwise of a secret message in the digital images. This is referred to as the 10 fold-cross validation. The 10 fold-cross validation methodology is implemented for the simulation of this study. In the method, the dataset is randomly apportioned into 10 equivalent sub-samples out of which one subsample serves as the validation set for testing the model. The three learner algorithms are treated same way and the cross validation is repeated in 10 clocks. The 10 results from the folds is subsequently averaged to generate a distinct estimation for each of the learner algorithms. This approach guarantees [33] that extracted feature vectors for each 8332 image instances are deployed for both training and validation while each subset is used for validation once each.

- RF depicts an ensemble of decision trees with the bagging methodology [34]. It identifies a threshold value, the binary class and the feature vector. Each node of the tree and its threshold are chosen to identify minimal variety possibilities in the children node. The diversity is measured by Gini criterion which ensures continuation of subdivisions until each node at the bottom of the tree is representative of one unique class.

- SVM is regarded as an extended version of maximum margin classifier and is most applicable where there are overlapping classes and for a non-linear decision [35]. The Kernel function is a tuning parameter which removes the computational requirements for a higher dimensional vector space. A polynomial function kernel uses a degree 2 to identify non-linear data through transformation into higher dimensions.

- The MLP is in the feed forward subcategory of artificial neural networks that is made up of at least three layers of input, hidden and output layers. Its multiple layers with the feature of non-linear activation is what distinguishes MLP from a linear perceptron, which is reputable for distinguishing and classifying non-linearly separable dataset [36]. The learning occurs in the perceptron by altering connection weights immediately each data points are processed which is based on the amount of error in the output comparable to the expected result.

TABLE 1: ATTRIBUTES OF DEEP EMBEDDERS

S/N	Pre-trained deep embedder	Size of extracted image descriptor from each image signal	Eventual training set for machine learning
1.	SqueezeNet	1000	1000/8332
2.	Inception v3	2048	2048/8332
3.	VGG16	4096	4096/8332
4.	VGG19	4096	4096/8332
5.	Painters	2048	2048/8332

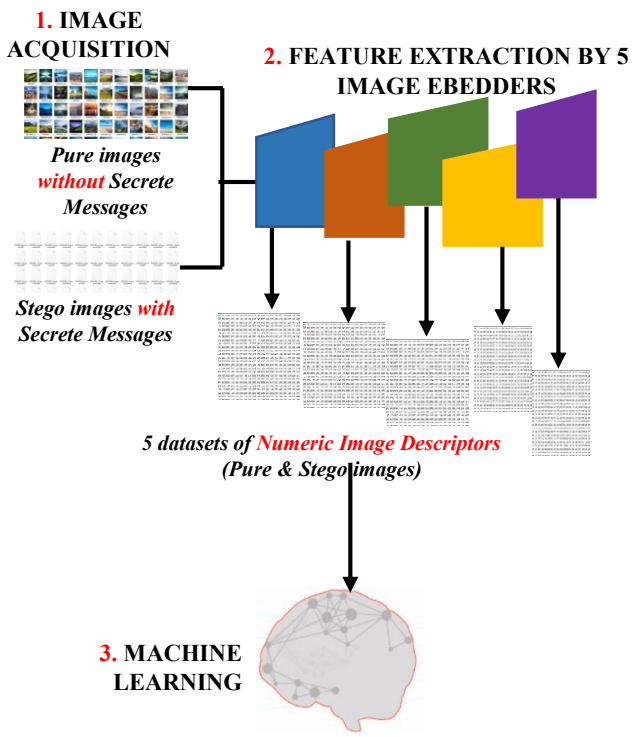

Figure 1: The conceptual framework of the study

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IV. RESULT AND DISCUSSION

The 10-fold cross-validation methodology of machine learning encapsulates the training (pattern recognition) and testing (steganography detection) in its framework described earlier in Figure 1. The result of the study is discussed in two folds highlighting the performance of the machine learning models and the contribution of the image embedders towards achieving experimental result. The performance metrics of the three learner algorithms on each of the five image embedders are presented in Table 2, Table 3, Table 4, Table 5, and Table 6. The Linear projection plot on each of the numeric descriptor set as extracted by the image embedders are likewise presented in Figure 3, Figure 4, Figure 5, Figure 6, and Figure 7. Experimental result RF outperformed SVM and MLP across the five datasets with respect to AUC performance metrics. The highest AUC weighted average is recorded with RF algorithm and Painters feature vector pair presented in Table 2 with 0.999 weighted average, followed by the RF/SqueezeNet pair in Table 4. Table 6 indicates AUC weighted average of 0.997 for the RF VG16 pair while the RF/VG19 pair recorded an AUC of 0.991. Evaluating the performance metrics of other algorithms on their CA and F1 score, in the light of the AUC returns Random forest and SqueezeNet as the best pair as their performance outweighs other combinations on all evaluation metrics with respect to the detection of secret messages in digital images. Hence, it is safe to conclude that size of the numeric image descriptor has less significance on detecting stego with machine learning owing to the smaller training size of SqueezeNet compared to other feature extraction models (TABLE 1). As may be observed, the ratio of correct predictions to the size of input samples as indicated by the CA is appreciable across the various combinations. The linear projection of the binary class-labelled data (as either positive or negative) is presented to show the internal workings of the numeric image descriptors towards the detection of secret messages in digital images. The attribute features 1, 2, and 3 out of the entire 1000 for SqueezeNet were the most significant for steganalysis just as VG19 returned only three feature attributes as the most significant unlike the other three embedders with five prominent discriminative feature attributes. The plots are indicative of the centrality of all image description data points with regards to the significant feature attributes.

TABLE 2: WEIGHTED AVERAGES OF MODELS ON PAINTERS IMAGE DESCRIPTORS

EMBEDDER	MODEL	AUC	CA	F1
Painters	RF	0.999	0.978	0.978
	SVM	0.989	0.967	0.967
	MLP	0.899	0.988	0.988

TABLE 3: WEIGHTED AVERAGES OF MODELS ON INCEPTIONV3 IMAGE DESCRIPTORS

EMBEDDER	MODEL	AUC	CA	F1
Inceptionv3	RF	0.992	0.978	0.978
	SVM	0.990	0.947	0.947
	MLP	0.991	0.962	0.962

TABLE 4: WEIGHTED AVERAGES OF MODELS ON SZUEEZENET IMAGE DESCRIPTORS

EMBEDDER	MODEL	AUC	CA	F1
SqueezeNet	RF	0.998	0.907	0.907
	SVM	0.989	0.982	0.982
	MLP	0.997	0.973	0.973

TABLE 5: WEIGHTED AVERAGES OF MODELS ON VGG19 IMAGE DESCRIPTORS

EMBEDDER	MODEL	AUC	CA	F1
VG19	RF	0.991	0.991	0.991
	SVM	0.990	0.989	0.989
	MLP	0.991	0.990	0.990

TABLE 6: WEIGHTED AVERAGES OF MODELS ON VGG16 IMAGE DESCRIPTORS

EMBEDDER	Model	AUC	CA	F1
VG16	RF	0.997	0.998	0.998
	SVM	0.988	0.974	0.974
	MLP	0.989	0.998	0.998

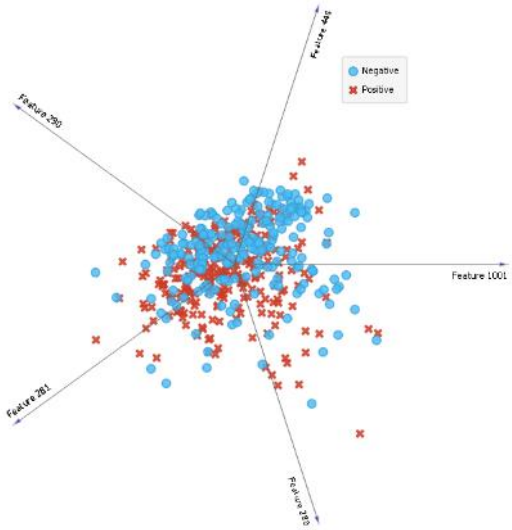

Figure 3: Linear projection plot for Painters

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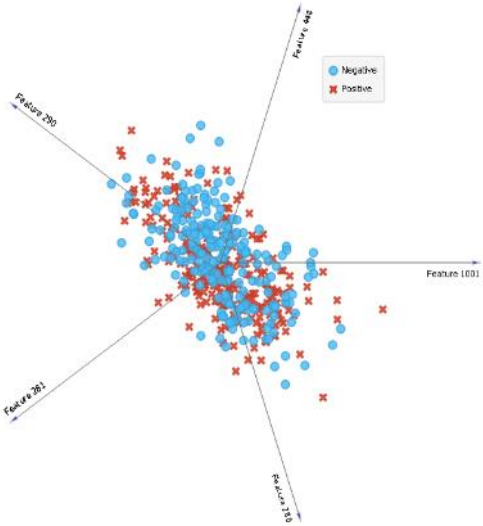

Figure 4: Linear projection plot for Inception

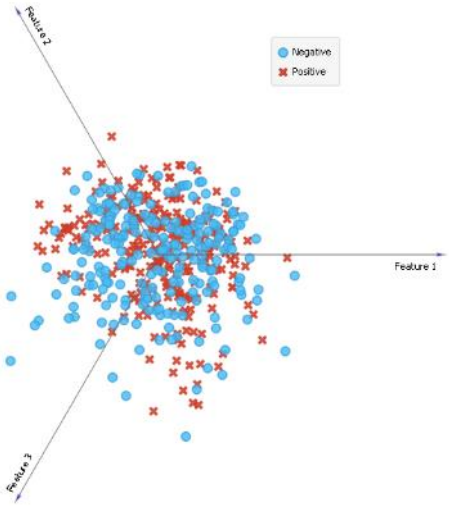

Figure 5: Linear projection plot for Squeezenet

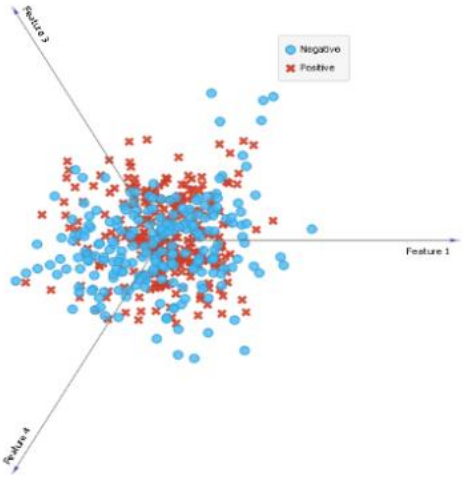

Figure 6: Linear projection plot for VG19

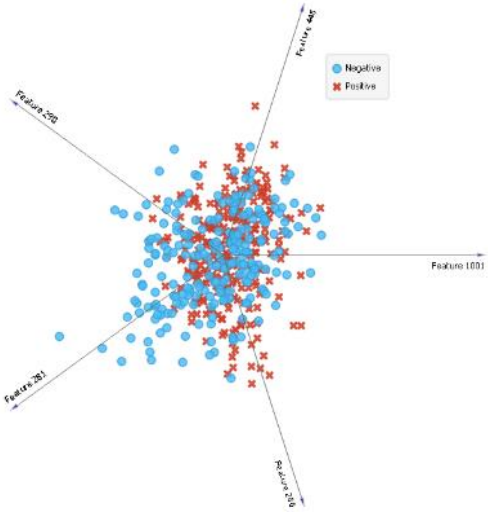

Figure 7: Linear projection plot for VG16

V. CONCLUSION AND RECOMMENDATION

Five image embedding networks are evaluated with three machine learning algorithms in this study to establish their discriminative ability towards detecting the presence of secret messages in digital images. The image embedding networks are used for extracting numeric vectors from all image inputs which serve as the input into the supervised machine learning algorithms. The input messages are of binary classes of *positive* (which is indicative of the presence of a secret message) and *negative* (which are clean images without secret messages). A 10 fold cross-validation methodology is employed for the machine learning phase of the study as experimental result returns Random forest with SqueezeNet embedder as the best ensemble for steganalysis. The result is indicative of the fact that the size of a training set for a machine learning-based steganalysis does not necessarily influence the performance of the learner algorithm as against what obtains in other research domains that employs machine learning. Future studies could employ synthetic minority oversampling on extracted image descriptors to further ascertain the performance of the Random forest and SqueezeNet ensemble for steganalysis.

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